

Error de series alternadas

∞

$\sum_{n=1}^{\infty} \frac{(-1)^n \cdot 2^n}{n!}$, que el error sea menor que $10^{-3} = 0,001$

$$\begin{array}{ccc} N & N+1 & \frac{2^n}{n!} < 10^{-3} \\ 2 & 3 & \end{array}$$

$$\rightarrow \frac{2^3}{3!} = \frac{8}{6} \neq 10^{-3}$$

$$6 \quad 7 \rightarrow \frac{2^7}{7!} \approx 0,025 \neq 0,001$$

$$9 \quad 10 \rightarrow \frac{2^{10}}{10!} \approx 0,0002 < 0,001$$

¿Cuál es el menor n que cumple?
 $N=9$

Calcule la suma

$$\sum_{x=1}^9 \frac{(-1)^x \cdot 2^x}{x!} = \boxed{-0,8679}$$

17. Considere la serie dada por $S = \sum_{n=1}^{\infty} (-1)^{n+1} \cdot \frac{\ln(n)}{n}$

b) Determine el menor valor de N de manera que la suma parcial S_N aproxime el valor

de la suma de la serie S con un error tal, que $|E_n| \leq 10^{-1}$

N	$N+1$	$\frac{\ln(n)}{n}$	$< 0,1 = 10^{-1}$
Respuesta	Prueba		

15	16	$\frac{\ln(16)}{16} = 0,17 \not< 0,1$
----	----	---------------------------------------

30	31	$\frac{\ln(31)}{31} = 0,11 \not< 0,1$
----	----	---------------------------------------

35	36	$\frac{\ln(36)}{36} = 0,09 < 0,1$
----	----	-----------------------------------

$N = 35$

$$\sum_{n=0}^{\infty} (-1)^{n+1} \cdot \frac{3^{n+1}}{(2n+1)!}$$

N	$N+1$	$\frac{3^{n+1}}{(2n+1)!}$	$< 10^{-8}$ 0,00000001
-----	-------	---------------------------	---------------------------

4	5	$\frac{3^{5+1}}{(2 \cdot 5+1)!} = 0,00001 \not<$	
---	---	--	--

6	7	$\frac{3^{7+1}}{(2 \cdot 7+1)!} = 0,00000001 <$
---	---	---

$N = 6$

18. Considere la serie dada por $\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$

b) Determine el menor número de términos necesarios que se deben tomar para aproximar la suma de la serie con un error menor que 0,0001

$$N \quad N+1 \quad \frac{1}{N!} < 0,0001$$

$$5 \quad 6 \quad \frac{1}{6!} \neq 0,0001 \neq 0,0001$$

$$6 \quad 7 \quad \frac{1}{7!} = 0,0001 \neq 0,0001$$

$$7 \quad 8 \quad \frac{1}{8!} = 0,00001 < 0,0001$$

Resp $N=7$ $\leftarrow$ Cantidad de terminos

c) Aproxime la suma de la serie usando el inciso b

R/ 0,3678

$$\sum_{x=0}^7 \frac{(-1)^x}{x!} = 0,3678$$

Criterio de la divergencia

Si $\sum a_n$ es convergente

entonces $\lim_{n \rightarrow \infty} a_n \neq 0$ Diverge

Si $a_n = 0$ no se sabe nada

∞

$$\sum_{n=1}^{\infty} \frac{-n}{2n+5}$$

$$\lim_{n \rightarrow \infty} \frac{-\cancel{n}}{2\cancel{n}} = \frac{-1}{2} \neq 0$$

Diverge

∞

$$\sum_{n=1}^{\infty} n^2$$

$$n \geq 1$$

$$\lim_{n \rightarrow \infty} n^2 = +\infty \neq 0 \therefore \text{Diverge}$$

Ojo!!!! NO usen este criterio
con series $\sum \left(\frac{1}{n^p} \right)$

$$13. \sum_{p=1}^{\infty} \frac{1}{\sqrt{p+4}}$$

$$\lim_{p \rightarrow \infty} \frac{1}{\sqrt{p+4}} = \frac{1}{+\infty} = 0$$

No se sabe nada

Criterio de comparación directa

$$\text{Sean } \sum_{n=p}^{\infty} a_n \text{ y } \sum_{n=p}^{\infty} b_n$$

$$\text{Con } L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

Si $L \neq 0$ y $L \neq +\infty$ a_n converge o diverge dependiendo
 $L = l, l \neq 0, l \neq +\infty$ si b_n converge o diverge

Si $L = 0$ y b_n converge,
 entonces a_n converge

Si $L = +\infty$ y b_n diverge
 entonces a_n diverge

$$\sqrt{\frac{L - \ln(h)}{5h^3 + 3h^2}} \leftarrow \frac{1}{h} \quad a_n, b_n?$$

$$\infty \quad \sum_{n=1}^{\infty} \sqrt{\frac{L - \ln(h)}{5h^3 + 3h^2}} \approx \sqrt{\frac{\cancel{h}}{5\cancel{h}^3}} = \frac{1}{\sqrt{h^2}} \approx \frac{1}{h} \quad \begin{matrix} b_n \\ \downarrow \end{matrix}$$

$$\infty \quad \sum_{n=1}^{\infty} \frac{1}{h} \quad \text{p-series, } p \leq 1 \quad \lim_{n \rightarrow +\infty} \frac{a_n}{b_n}$$

Diverge

$$\lim_{n \rightarrow +\infty} \frac{\sqrt{\frac{L - \ln(h)}{5h^3 + 3h^2}}}{\frac{1}{h}}$$

$$\lim_{n \rightarrow +\infty} \frac{\sqrt{\frac{\cancel{h}}{5\cancel{h}^3}}}{\frac{1}{h}} \rightarrow \frac{1}{\sqrt{h^2}} = \frac{1}{h}$$

$$\lim_{n \rightarrow +\infty} \frac{\cancel{1}}{\sqrt{5} \cdot \cancel{h}} \cdot \frac{\cancel{1}}{\cancel{h}}$$

$$\lim_{n \rightarrow +\infty} \frac{1}{\sqrt{5}}$$

$$\infty \quad \sum_{n=1}^{\infty} \sqrt{\frac{L - \ln(h)}{5h^3 + 3h^2}}$$

b_n diverge $\checkmark$
 $L = K, K \neq 0 \wedge K \neq +\infty$
 $\therefore a_n$ diverge

∞

$$\sum_{n=1}^{\infty} \frac{1}{n^3 \sqrt[3]{2+\ln(n)}} \approx \frac{1}{n^3 \sqrt[3]{\ln(n)}}$$

∞

$$\sum_{n=1}^{\infty} \frac{1}{n^3 \sqrt[3]{\ln(n)}}$$

$$f'(x) = \frac{1}{x^3 \sqrt[3]{\ln(x)}}$$

$$= \frac{1}{x \cdot \ln(x)^{\frac{7}{3}}}$$

$$= \frac{-1}{(x \ln(x)^{\frac{7}{3}})^2} \cdot \ln(x)^{\frac{7}{3}} + \cancel{x} \cdot \frac{7}{3} \ln(x)^{\frac{7}{3}-1} \cdot \frac{1}{x}$$

$$= \frac{-\ln(x)^{\frac{7}{3}} + \frac{7}{3} \ln(x)}{(x \ln(x)^{\frac{7}{3}})^2} \quad \therefore \text{Decrease}$$

$$\int_1^{+\infty} \frac{1}{x^3 \sqrt[3]{\ln(x)}}$$

$$\int \frac{1}{x \cdot \ln(x)^{\frac{7}{3}}} dx \quad u = \ln(x) \\ du = \frac{1}{x}$$

$$\int \frac{1}{u^{\frac{7}{3}}} \rightarrow \int u^{-\frac{7}{3}} \rightarrow \frac{u^{-\frac{2}{3}}}{-\frac{2}{3}}$$

$$\rightarrow \frac{\ln^{\frac{2}{3}}(x)}{\frac{2}{3}} \quad \begin{array}{c} +\infty \\ | \\ 1 \end{array}$$

$$\lim_{x \rightarrow +\infty} \frac{\ln^{\frac{2}{3}}(x)}{\frac{2}{3}} - \frac{\ln^{\frac{2}{3}}(1)}{\frac{2}{3}}$$

$$+\infty - \quad \nearrow$$

$$+\infty$$

b_n diverge

$$\lim_{n \rightarrow +\infty} \frac{\frac{1}{n^3 \sqrt[3]{2+\ln(n)}}}{\frac{1}{n^3 \sqrt[3]{\ln(n)}}}$$

$$\lim_{n \rightarrow +\infty} \frac{\frac{1}{n^3 \sqrt[3]{\ln(n)}}}{\frac{1}{n^3 \sqrt[3]{\ln(n)}}} = 1$$

$$\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = 1, \neq 0, \neq +\infty$$

$\nearrow b_n$ diverge

$$\therefore \lim_{n \rightarrow +\infty} \frac{1}{n^3 \sqrt[3]{2+\ln(n)}} \text{ diverge}$$

2. Analice y justifique la convergencia de la siguiente serie, y calcule su suma

$$\sum_{k=3}^{\infty} \frac{9 \cdot 5^{-k}}{(-3)^{2-k}} - \frac{2}{k^2 - 1} \quad n=k$$

$$\sum_{n=3}^{\infty} \frac{9 \cdot 5^{-n}}{(-3)^{2-n}} \quad - \quad \sum_{n=3}^{\infty} \frac{2}{n^2 - 1}$$

$$\sum_{n=3}^{\infty} \frac{9 \cdot (5^{-1})^n}{(-3)^2 \cdot (-3^{-1})^n} \quad \sum_{n=3}^{\infty} \frac{2}{(n-1)(n+1)}$$

$\frac{a^2 - b^2}{(a-b)(a+b)}$

$$\sum_{n=3}^{\infty} \left(\frac{5^{-1}}{(-3)^{-1}} \right)^n \quad \frac{2}{(n-1)(n+1)} = \frac{A}{n-1} + \frac{B}{n+1}$$

$$\sum_{n=3}^{\infty} \left[\left(\frac{5}{-3} \right)^{-1} \right]^n$$

$$2 = A(n+1) + B(n-1)$$

$$\begin{matrix} n+1=0 & n-1=0 \\ n=-1 & n=1 \end{matrix}$$

$$\sum_{n=3}^{\infty} \left(\frac{-3}{5} \right)^n$$

$$n=1 \rightarrow 2 = A(1+1) + B(1-1)$$

$$2 = 2A$$

$$\boxed{A=1}$$

$$|r| = \left| \frac{-3}{5} \right| = \frac{3}{5} < 1$$

$\therefore$ Converge

$$n=-1 \rightarrow 2 = A(-1+1) + B(-1-1)$$

$$2 = -2B$$

$$\boxed{B=-1}$$

$$\frac{\left(\frac{-3}{5} \right)^3}{1 - \frac{-3}{5}} \quad \frac{r^p}{1-r} \quad \frac{2}{(n-1)(n+1)} = \frac{A}{n-1} + \frac{B}{n+1}$$

$$\boxed{\frac{-27}{200}}$$

$$= \frac{1}{n-1} - \frac{1}{n+1}$$

$$\sum_{n=3}^{\infty} \frac{1}{n-1} - \frac{1}{n+1}$$

$\downarrow_n$ $\downarrow_{n+1}$ $\frac{(n+1)-1}{n} = \frac{1}{n}$

~~X~~

$$\sum_{n=3}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) + \sum_{n=3}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

$$\frac{1}{3-1} - \lim_{n \rightarrow \infty} \frac{1}{n} + \frac{1}{3} - \lim_{n \rightarrow \infty} \frac{1}{n+1}$$

$$\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

$$\frac{-27}{200} + \frac{5}{6}$$

$$\frac{739}{600}$$

$$1. \sum_{n=2}^{\infty} \frac{(-1)^n \cdot (n^2 - n) \cdot 3^n + 5^n}{5^n \cdot (n^2 - n)}$$

$$(r)^h$$

↓
IT

$$a_h^h$$

|
Expression

∞

$$\sum_{h=2}^{\infty} \frac{(-1)^h \cdot \cancel{(h^2 - h)} \cdot 3^h}{5^h \cdot \cancel{(h^2 - h)}} + \frac{\cancel{5^h}}{\cancel{5^h (h^2 - h)}}$$

∞

$$\sum_{h=2}^{\infty} \frac{(-1)^h (3)^h}{(5)^h} + \frac{1}{h^2 - h}$$

∞

$$\sum_{h=2}^{\infty} \left(\frac{-1 \cdot 3}{5} \right)^h + \frac{1}{h(h-1)}$$

∞

$$\sum_{h=2}^{\infty} \left(\frac{-3}{5} \right)^h + \sum_{h=2}^{\infty} \frac{1}{h(h-1)}$$

$$\frac{\left(\frac{-3}{5}\right)^2}{1 - \frac{-3}{5}} \quad \text{red } r^p \quad \frac{1}{h(h-1)} = \frac{A}{h-1} + \frac{B}{h}$$

$h-1=0 \quad h=0$
 $h=1$

9
70

$$1 = A(h) + B(h-1)$$

$$h=1 \rightarrow 1 = A$$

$$A=1$$

$$h=0 \rightarrow 1 = -B$$

$$B=-1$$

$$\Rightarrow \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$$

$$\begin{matrix} \infty \\ \varepsilon \end{matrix} \quad \frac{1}{n-1} - \frac{1}{n} \quad \frac{Cn + 1 - 1}{n}$$

$$\frac{1}{2-1} - \lim_{n \rightarrow \infty} \frac{1}{n}$$

$$\frac{1}{1} = 1$$

$$\frac{9}{70} + 1$$

$\frac{79}{70}$
